

Assignment 01

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MA511

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Notation 1. We write $\mathbb{N}$ for the set of naturals, which we only use for convenience of notation, since sets have not been formally introduced. Given $n \in \mathbb{N}$, we write $s(n)$ for the successor of n . Multiplication is always denoted by juxtaposition or $\cdot$. References to in-class results/definitions are either given by name, or cited in the first lecture notes [Enk26].

Problem 1 (Addition is commutative). *For any naturals n, m , we have $n + m = m + n$.*

Proof. We proceed by induction on n . Fix a natural m , and let $n = 0$. By definition of addition, we have $0 + m = m$, and by [Enk26] §2.2 Lemma 1, we have $m + 0 = m$. Thus, $m + 0 = m = 0 + m$.

Now assume $n + m = m + n$ for some natural n , and consider the expression $s(n) + m$. By definition of addition, $s(n) + m = s(n + m)$. By the inductive hypothesis, $n + m = m + n$, and thus $s(n + m) = s(m + n)$. By [Enk26] §2.2 Lemma 2, we have $s(m + n) = m + s(n)$. Thus, we have equalities

$$s(n) + m = s(n + m) = s(m + n) = m + s(n).$$

The statement now follows by induction. ■

Problem 2 (Cancellation law). *Let a, b, c be naturals such that $a + b = a + c$. Then $b = c$.*

Proof. We proceed by induction on a . Let $a = 0$, and let $b, c \in \mathbb{N}$ such that $a + b = a + c$. Then by definition of addition, $0 + b = b$ and $0 + c = c$. Thus,

$$b = 0 + b = 0 + c = c.$$

Now assume $a + b = a + c$ implies that $b = c$ for some natural a and all naturals b, c , and suppose that $s(a) + b = s(a) + c$ for some $b, c \in \mathbb{N}$. Then by definition of addition, we have

$$s(a + b) = s(a) + b = s(a) + c = s(a + c).$$

By Peano axiom 4, it follows that $a + b = a + c$. By the inductive hypothesis, $b = c$. By the method of induction, we are done. ■

Problem 3 (Order is transitive). *Prove that ordering is transitive.*

Proof. Let a, b, c be naturals with $c \leq b \leq a$. We must show $c \leq a$. The following lemma is required.

Lemma 3.1. *Let a, b, c be naturals. Then $(a + b) + c = a + (b + c)$.*

Proof. We proceed by induction on a . Fix naturals b, c , and let $a = 0$, and recall that $0 + b = b$ and $0 + (b + c) = b + c$ by definition of addition. Thus

$$(0 + b) + c = b + c = 0 + (b + c).$$

Now let a be an arbitrary natural, and consider the successor $s(a)$. We have

$$\begin{aligned}
(s(a) + b) + c &= s(a + b) + c && \text{Definition of addition} \\
&= s((a + b) + c) && \text{Definition of addition} \\
&= s(a + (b + c)) && \text{Inductive hypothesis} \\
&= s(a) + (b + c) && \text{Definition of addition.}
\end{aligned}$$

By the method of induction, the statement is proven. ■

We now continue with problem 3. Since $c \leq b$, there is a natural n such that $c + n = b$. Since $b \leq a$, there is a natural m such that $b + m = a$. We have

$$a = b + m = (c + n) + m \stackrel{!}{=} c + (n + m),$$

where the equality marked with a $!$ follows by lemma 3.1. Since $n + m$ is a natural, by definition of $\leq$, we have $c \leq a$. ■

Problem 4 (Multiplication is commutative). *Let n, m be naturals. Then $nm = mn$.*

Proof. We use $ab = ba$ instead of $nm = mn$, to improve readability. We proceed by induction on a . Let $a = 0$, and by definition of multiplication, $0b = 0$ for all $b \in \mathbb{N}$. We must show $b0 = 0$. We proceed by induction on b . When $b = 0$, we observe that $0 \cdot 0 = 0$ is true by definition of multiplication. Now assume $b0 = 0$, and consider the expression $s(b)0$. By definition of multiplication, $s(b)0 = (b0) + 0$. By the inductive hypothesis, $b0 = 0$, and by definition of addition, $0 + 0 = 0$. Thus,

$$s(b)0 = (b0) + 0 = 0 + 0 = 0.$$

Thus, $b0 = 0$ holds for all b .

Now we assume the inductive hypothesis for a ; that is, we assume $ab = ba$ for some arbitrary natural a and all naturals b . By definition of multiplication,

$$s(a)b = ab + b = ba + b.$$

It suffices to show $bs(a) = ba + b$ for all naturals b . We proceed by induction on b . First, take $b = 0$. Notice that $0s(a) = 0$ by definition of multiplication, and $0a + 0 = 0 + 0 = 0$ by definition of multiplication and addition. Now assume the statement holds for b an arbitrary natural. We want to show $s(b)s(a) = s(b)a + s(b)$. We have

$$\begin{aligned}
s(b)s(a) &= bs(a) + s(a) && \text{Definition of multiplication} \\
&= (ba + b) + s(a) && \text{Inductive hypothesis} \\
&= ba + (b + s(a)) && \text{Lemma 3.1} \\
&= ba + (s(a) + b) && \text{Problem 1} \\
&= ba + s(a + b) && \text{Definition of addition} \\
&= ba + s(b + a) && \text{Problem 1} \\
&= ba + (s(b) + a) && \text{Definition of addition} \\
&= ba + (a + s(b)) && \text{Problem 1} \\
&= (ba + a) + s(b) && \text{Lemma 3.1} \\
&= s(b)a + s(b) && \text{Definition of multiplication.}
\end{aligned}$$

Thus, the statement is proven. ■

Problem 5 ($\mathbb{N}$ has no zero-divisors). *Let n, m be naturals. Then $nm = 0$ if and only if at least one of n, m is 0.*

We once again use a, b in place of n, m to improve readability. The proof of one direction requires a lemma, which was proven in class but is not contained in the notes.

Lemma 5.1. *Let a be a natural. Then a is either 0, or a successor.*

Proof. We proceed by induction. If $a = 0$, then $a = 0$, and by Peano axiom 2 a is not a successor. Now let a be an arbitrary natural. Then $s(a)$ is a successor. Moreover, by Peano axiom 2, $s(a) \neq 0$. The statement is now proven. ■

Proof of problem 5. ($\implies$) Let $ab = 0$. If $a = 0$ or $b = 0$ we are done, so suppose for contradiction that $a, b \neq 0$. Applying lemma 5.1 to $a, b \neq 0$, we see that $a = s(\tilde{a})$ and $b = s(\tilde{b})$ for some naturals $\tilde{a}, \tilde{b}$. By the definition of multiplication, we compute

$$ab = s(\tilde{a})s(\tilde{b}) = \tilde{a}s(\tilde{b}) + s(\tilde{b}).$$

Applying problem 4 and the definition of multiplication, we have

$$\tilde{a}s(\tilde{b}) + s(\tilde{b}) = s(\tilde{b})\tilde{a} + s(\tilde{b}) = (\tilde{b}\tilde{a} + \tilde{a}) + s(\tilde{b}).$$

We now have

$$\begin{aligned} (\tilde{b}\tilde{a} + \tilde{a}) + s(\tilde{b}) &= s(\tilde{b}) + (\tilde{b}\tilde{a} + \tilde{a}) && \text{Problem 1} \\ &= s(\tilde{b} + (\tilde{b}\tilde{a} + \tilde{a})) && \text{Definition of addition.} \end{aligned}$$

Notice that the equation is thus equal to a successor, and by Peano axiom 2, all successors are nonzero. Thus,

$$ab = s(\tilde{b} + (\tilde{b}\tilde{a} + \tilde{a})) \neq 0,$$

and the first direction is proven.

($\impliedby$) Let at least one of a, b be 0. By problem 4, we can take $a = 0$ without loss of generality. Then by definition of multiplication,

$$ab = 0b = 0.$$

The statement is proven. ■

Problem 6. *For any naturals a, b, c , we have*

$$a(b + c) = ab + ac \tag{1}$$

and

$$(b + c)a = ba + ca. \tag{2}$$

Proof. Assume (1) holds. Then (2) follows by problem 4:

$$(b + c)a = a(b + c) = ab + ac = ba + ca.$$

Thus, it suffices to prove (1). We proceed by induction on a . If $a = 0$, then $0(b + c) = 0$ by definition of multiplication. Additionally, $0b = 0c = 0$ by definition of multiplication, and thus $0b + 0c = 0 + 0 = 0$ by definition of addition. Thus, $0(b + c) = 0b + 0c$.

Now assume $a(b + c) = ab + ac$ for some arbitrary natural a . We have

$$\begin{aligned}
 s(a)(b + c) &= a(b + c) + (b + c) && \text{Definition of multiplication} \\
 &= (ab + ac) + (b + c) && \text{Inductive hypothesis} \\
 &= (ab + b) + (ac + c) && \text{Repeated applications of problem 1 and lemma 3.1} \\
 &= s(a)b + s(a)c && \text{Definition of multiplication.}
 \end{aligned}$$

By the method of induction, we are done. ■

References

[Enk26] Thomas Enkowsky. Ma511 notes, class 1. Lecture notes, 2026.